

Paper Practice 3.2**Functions**

Show all of your work in the space provided. For every yes-or-no question, explain your reasoning in a complete sentence.

- 1.** Determine whether each relation is a function. Explain. (Relations)

a. $\{(2, 5), (4, 7), (8, 11), (4, 13)\}$

b. $\{(-3, 6), (0, 6), (5, 6), (9, 6)\}$

c. $\{(1, 2), (2, 4), (3, 6), (2, 8)\}$

- 2.** State the domain and the range of each relation. (Domain & Range)

a. $\{(-4, 3), (-1, 0), (2, 5), (6, 3)\}$

b. $\{(0, -2), (1, -2), (2, -2)\}$

- 3.** A mapping pairs $3 \rightarrow 7$, $5 \rightarrow 7$, $8 \rightarrow 9$, and $3 \rightarrow 11$. Is the relation a function? Explain. (Mapping)

- 4.** If $f(x) = 3x - 7$, find each value. (Function Notation)

a. $f(4)$

b. $f(-2)$

c. $f(0)$

- 5.** If $g(x) = x^2 + 2x$, find each value. (Function Notation)

a. $g(3)$

b. $g(-5)$

c. $g(0)$

- 6.** If $f(x) = 2x + 9$, find the value of x for which $f(x) = 21$. (Solve for the Input)

- 7.** A vertical line drawn at $x = 2$ crosses a graph at $(2, 1)$ and at $(2, -3)$. Is the graph a function? Explain. (Vertical Line Test)

- 8.** Madison took five quizzes. Her scores were Quiz 1: 86, Quiz 2: 92, Quiz 3: 78, Quiz 4: 86, Quiz 5: 84. (Analyze Data)

a. What is the domain?

b. What is the range?

c. Is this relation a function? Explain.

- 9.** The table below shows a relation. Is it a function? Explain. (Table)

a. x : $-2, -1, 0, 1$ y : $5, 5, 5, 5$

- 10.** Error Analysis – Gus says $\{(1, 4), (2, 4), (3, 4)\}$ is not a function because the value 4 repeats. Identify his mistake, then state whether the relation is a function. (Error Analysis)

- 11.** Write the relation $\{(1, 3), (2, 6), (3, 9), (4, 12)\}$ as a rule in function notation. (Write a Rule)

- 12.** Write a set of four ordered pairs that is NOT a function, then explain in one sentence what makes it fail. (Construct)